

WASP-43b

R-2283

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_150228 · created 2026-08-17T00:17:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457081.79146 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.03
Transit depth (Rp/Rs)^2	1.51 +/- 0.75 [%]
Orbital Inclination (inc)	83.17 +/- 1.98
Airmass coefficient 1 (a1)	1.143 +/- 0.016
Airmass coefficient 2 (a2)	-0.095 +/- 0.069
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#3 - [55, 268]
Optimal Aperture	3.5
Optimal Annulus	15.47
Transit Duration (day)	0.0505 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.03 vs 0.1594 (deviation 1.16σ)
Duration fit vs published	0.0505 d vs 0.0512 d (deviation 0.11σ)
Mid-time O–C	1.1 min
Depth SNR	3.9
Residual scatter	1.42 %

Light curve

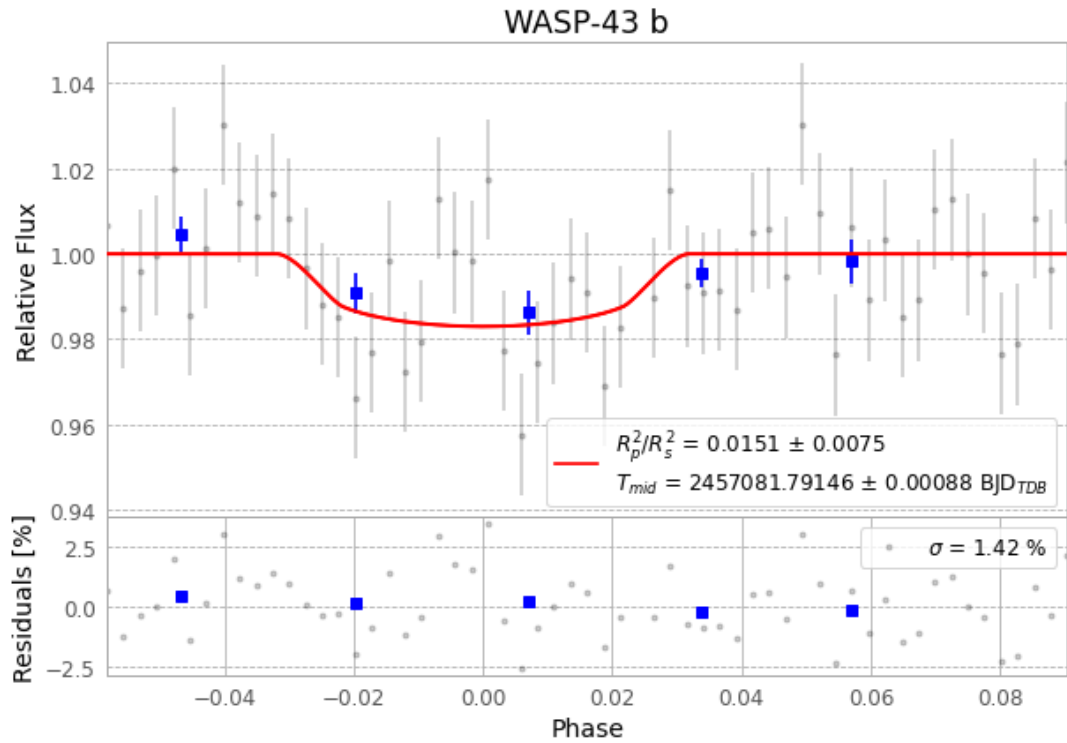


Figure 1 — FinalLightCurve_WASP-43 b_2015-02-27.png (SHA-256 3c10cfe5adb7ea5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [186, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 abf32383749f8e3d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

59 FITS frames (39 MB), WASP-43150228054212.FITS → WASP-43150228083612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150228.log (869cec22dc10...), FinalLightCurve_WASP-43 b_2015-02-27.png (3c10cfe5adb7...), FinalLightCurve_WASP-43 b_2015-02-27.csv (297ad4686bd8...)

Compute resources

Wall time 143.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:17Z.